

Variance Risk Premium Decomposition and Regime-Conditional Harvesting

A Dual-Market Empirical Study across SPX/VIX in the US and NIFTY/India VIX in India

EPAT Final Project Report

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Programme: Executive Programme in Algorithmic Trading

Project type: Empirical volatility research, regime modelling, and research-layer strategy evaluation

Report source: reports/final/final_report.md

PDF export: reports/final/final_report.pdf

Report Control Statement

This report is the final Markdown source for the EPAT submission package.

The project is a public-data empirical research project. It measures and studies the variance risk premium across the US and Indian equity-index volatility markets, evaluates regime-conditioned research-proxy harvesting rules, and documents a paper-signal readiness layer.

This report does not claim live-trading profitability, true option-chain PnL, account returns, broker execution, or causal cross-market transmission. Generated data panels remain local and are not committed to GitHub.

Every major result claim in this report must be traceable to:

```
reports/final/result_claims_audit.md
```

Every table and figure used in this report must be listed in:

```
reports/final/table_inventory.md
reports/final/figure_inventory.md
reports/final/selected_artifacts.md
```

Abstract

The variance risk premium is the empirical spread between implied variance and subsequently realised variance. In equity-index markets, implied volatility indices such as VIX and India VIX often stand above subsequent realised volatility, but unconditional short-volatility exposure can suffer severe losses during stress periods.

This project builds a reproducible dual-market research pipeline for the US and India. The US leg uses SPX/SPY and VIX. The India leg uses NIFTY 50 and India VIX. The pipeline constructs realised variance from daily OHLC data, converts implied-volatility indices into annualised implied variance proxies, builds variance risk premium features and forward outcome labels, estimates HAR-RV forecasts, fits threshold, Gaussian HMM, Markov autoregression, and MSVOL-style robustness regime models, constructs regime-conditioned exposure intentions, evaluates a vectorised research-proxy backtest, and performs cross-market US-India diagnostics.

The empirical objective is not to prove live tradability. It is to test whether regime-conditioned exposure improves research-proxy risk-adjusted behaviour relative to unconditional short-volatility harvesting, and whether US and Indian VRP regimes display useful descriptive or predictive relationships. Phase 11 adds a paper-signal readiness appendix with live-order guards, but no broker orders are placed. Phase 12 paper execution is intentionally skipped and left as future optional work.

Executive Summary

This project studies whether variance risk premium harvesting can be made more defensible by conditioning exposure on volatility regimes.

The research design has three central components:

- **VRP measurement:** estimate realised variance from daily OHLC data and compare it with implied variance proxies derived from VIX and India VIX.
- **Regime decomposition:** classify market states using threshold regimes, Gaussian HMM, Markov autoregression, and MSVOL robustness diagnostics.
- **Research-proxy strategy evaluation:** compare unconditional short-volatility harvesting with regime-conditioned exposure intentions in a vectorised backtest.

The project is dual-market by design. It evaluates the US and Indian markets separately, then adds a cross-market analysis layer to study same-date co-movement, lagged-US predictive diagnostics, and an analysis-only India overlay.

Main evidence categories:

Evidence area	Primary source
Data and VRP construction	reports/tables/vrp_summary.csv; reports/tables/vrp_metadata.json
HAR forecast layer	reports/tables/har_forecast_accuracy.csv; reports/tables/har_no_lookahead_audit.csv
Regime model ladder	Phase 5, 6, 7, and 8 diagnostic tables
Strategy signal construction	reports/tables/phase_9/strategy_signal_summary.csv
Research-proxy backtest	reports/tables/phase_10/backtest_summary.csv
Robustness checks	reports/tables/phase_10/robustness_cost_sensitivity.csv; reports/tables/phase_10/robustness_subperiods.csv
Paper-signal readiness	reports/tables/phase_11/live_order_guard_report.json
Cross-market analysis	reports/tables/phase_13/logistic_model_comparison.csv; reports/tables/phase_13/lead_lag_table.csv

Numeric findings are intentionally left as placeholders until the final local artifacts are inspected.

1. Motivation

Variance risk premium research sits between market microstructure, volatility forecasting, and systematic risk-premia design.

Equity-index options often embed a premium for protection demand, crash risk, jump risk, and volatility uncertainty. Short-volatility strategies attempt to harvest that premium, but unconditional exposure is structurally vulnerable to volatility spikes. The central motivation of this project is to evaluate whether regime conditioning can retain useful premium exposure while reducing exposure during historically adverse states.

The project is designed for EPAT as both a research and implementation exercise:

- It uses public daily data rather than private option-chain data.
 - It separates research-proxy evaluation from executable trading.
 - It enforces no-lookahead constraints.
 - It compares two markets with different structures.
 - It adds a paper-signal readiness appendix without crossing into broker execution.
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2. Research Questions

The project is built around the following questions.

RQ1 — VRP existence and construction

Can VIX and India VIX be used as implied-volatility proxies to construct a consistent public-data VRP panel for the US and Indian markets?

RQ2 — Forecasting layer

Does a HAR-RV model provide a useful prospective realised-variance forecast layer for constructing HAR-based VRP features under point-in-time constraints?

RQ3 — Regime decomposition

Do threshold, Gaussian HMM, and Markov autoregression models produce economically interpretable volatility or VRP regimes?

RQ4 — Regime-conditioned harvesting

Do regime-conditioned exposure rules improve research-proxy risk-adjusted behaviour relative to unconditional short-volatility harvesting in the tested sample?

RQ5 — Robustness

Are results sensitive to assumed transaction costs, subperiods, crisis windows, model choice, and sample alignment?

RQ6 — Cross-market structure

Do US and Indian VRP/regime diagnostics display descriptive co-movement or lagged predictive associations, without implying causal transmission?

RQ7 — Operational readiness

Can the final research signal be converted into a guarded paper-signal format without placing broker orders?

3. Data

3.1 Markets

Market	Underlying	Implied-volatility proxy
US	SPX / SPY	VIX
India	NIFTY 50	India VIX

3.2 Data sources

The project uses public daily market data. Data loaders support official and fallback sources where possible.

Primary data categories:

Data category	US	India
Underlying OHLC	SPX/SPY daily OHLC	NIFTY 50 daily OHLC
Implied-volatility proxy	VIX	India VIX
Frequency	Daily	Daily
Storage policy	Local generated artifacts	Local generated artifacts

3.3 Data caveats

The data layer has important limitations:

- VIX and India VIX are implied-volatility index proxies, not variance swap quotes.
- Daily OHLC data is not full intraday realised variance.
- Yahoo Finance data is convenient but not official exchange data.
- NSE scripted access may be blocked; manual CSV override is supported.
- Generated raw and processed data panels remain local and are not committed.

3.4 Data coverage

Local evidence inspection found one tracked data-audit summary row and a separate calendar-alignment table. The data-audit row covers the US VIX source, while the calendar mismatch table records aligned IV/RV ranges for both markets.

Data audit summary:

market	dataset	source	symbol	start_date	end_date	n_rows	n_missing_close
US	us_vix	fred_vixcls	VIXCLS	1990-01-02	2026-05-14	9186.0000	0.0000

Calendar alignment summary:

market	iv_start	iv_end	rv_start	rv_end	iv_rows	rv_rows	common_dates
US	1990-01-02	2026-05-14	1990-01-02	2026-05-15	9186.0000	9160.0000	9156.0000
INDIA	2009-03-02	2026-05-15	2007-09-17	2026-05-15	4220.0000	4576.0000	4204.0000

The calendar evidence should be read conservatively: the IV and RV calendars do not have identical raw date ranges, and final feature construction uses common dates where the required inputs are available.

4. Methodology Overview

The project follows this pipeline:

```
public daily data
↓
clean OHLC / VIX / India VIX series
↓
realised variance estimators
↓
implied variance construction
↓
variance risk premium construction
↓
HAR-RV forecast
↓
threshold regimes
```

↓
Gaussian HMM regimes
↓
Markov autoregression regimes
↓
MSVOL robustness appendix
↓
regime-conditioned strategy signals
↓
vectorised research backtest
↓
robustness checks
↓
IBKR paper-signal readiness appendix
↓
cross-market US-India analysis
↓
final report and release package

The design separates four layers:

Layer	Purpose	Output type
Data and feature layer	Build RV, IV, VRP, HAR-VRP features	Local panels and summary tables
Regime layer	Estimate market states	Regime probabilities and diagnostics
Strategy layer	Convert regimes into exposure intentions	Signal panels
Evaluation layer	Evaluate research-proxy behaviour	Backtest diagnostics and robustness tables

5. Realised Variance Construction

Realised variance is estimated from daily OHLC data. The project includes multiple estimators, with Garman-Klass 22-trading-day annualised realised variance used as the primary proxy.

The inspected metadata identifies:

Field	Value
primary_estimator	garman_klass

Field	Value
primary_column	rv_gk_22d_ann
rolling_convention	22 trading days
annualization_periods	252

Candidate estimators include close-to-close, Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang variants. These should all be read as daily-data realised-variance proxies rather than observed variance swap outcomes.

Primary realised-variance summary:

market	column	count	missing	mean	median	std	min	max
US	rv_gk_22d_ann	9139.0000	21.0000	0.0191	0.0108	0.0303	0.0013	0.4369
INDIA	rv_gk_22d_ann	4555.0000	21.0000	0.0299	0.0140	0.0545	0.0033	0.4845

Selected PDF candidate figures, not embedded in this pass:

Figure path	Caption constraint	Status
reports/figures/rv_estimators_us.png	Daily OHLC realised-variance estimator comparison, US	optional/maybe
reports/figures/rv_estimators_india.png	Daily OHLC realised-variance estimator comparison, India	optional/maybe

6. Implied Variance and VRP Construction

VIX and India VIX are converted into implied variance proxies by squaring the volatility index after percentage scaling.

Conceptually:

$$\text{implied_variance_proxy} = (\text{implied_volatility_index} / 100)^2$$

The inspected VRP metadata identifies `rv_gk_22d_ann` as the primary realised-variance column, `vrp_backward_gk` as the primary backward VRP column, and `vrp_forward_expost_gk_label` as the forward ex-post outcome label. The horizon is 22 trading days, with the metadata noting that 22 trading days approximate the 30-calendar-day VIX / India VIX horizon.

Compact VRP construction summary:

market	iv_ann_mean	iv_ann_median	rv_gk_22d_ann_lag1_mean	rv_gk_22d_ann_lag1_median	vrp_backward_gk_mean	vrp_backward_gk_median	vrp_forward_expost_gk_label_mean	vrp_forward_expost_gk_label_median
INDIA	0.0405	0.0281	0.0208	0.0128	0.0196	0.0145	0.0199	0.0148
US	0.0439	0.0310	0.0191	0.0108	0.0247	0.0184	0.0248	0.0185

The forward ex-post VRP label is an evaluation outcome only. It is not a tradable feature at signal time.

Selected final-report figures:

 US implied variance, realised variance, and VRP proxy construction diagnostic. VIX is an implied-volatility proxy, not a variance swap quote.

US implied variance, realised variance, and VRP proxy construction diagnostic. VIX is an implied-volatility proxy, not a variance swap quote.

 India implied variance, realised variance, and VRP proxy construction diagnostic. India VIX is an implied-volatility proxy, not a variance swap quote.

India implied variance, realised variance, and VRP proxy construction diagnostic. India VIX is an implied-volatility proxy, not a variance swap quote.

7. HAR-RV Forecasting

The HAR-RV layer estimates prospective realised variance using lagged realised-variance features. Its role is to provide a model-based expected realised variance input for prospective VRP construction.

Compact forecast comparison:

market	forecast_col	target_col	n_obs	rmse	mae	correlation
US	har_rv_gk_22d_forecast_ann	rv_gk_22d_forward_ann_label	8590.0000	0.0247	0.0100	0.6163
US	naive_lagged_22d_rv_ann	rv_gk_22d_forward_ann_label	9112.0000	0.0277	0.0108	0.5829
US	expanding_mean_forward_rv_baseline	rv_gk_22d_forward_ann_label	9089.0000	0.0305	0.0141	0.0515
US	rolling_mean_forward_rv_baseline	rv_gk_22d_forward_ann_label	9089.0000	0.0311	0.0148	0.1008
INDIA	har_rv_gk_22d_forecast_ann	rv_gk_22d_forward_ann_label	3638.0000	0.0326	0.0131	0.1391
INDIA	naive_lagged_22d_rv_ann	rv_gk_22d_forward_ann_label	4181.0000	0.0384	0.0126	0.3184
INDIA	expanding_mean_forward_rv_baseline	rv_gk_22d_forward_ann_label	4137.0000	0.0333	0.0182	0.2406
INDIA	rolling_mean_forward_rv_baseline	rv_gk_22d_forward_ann_label	4137.0000	0.0332	0.0159	0.2270

The HAR no-lookahead audit contains 13360 rows. It records 12228 rows with `forecast_available=True`; blocked rows are labelled with observed reasons such as `insufficient_training_history`, `missing_target_metadata`. The audit columns include `rule_target_end_before_forecast_date`, `forecast_available`, and `blocked_reason`, supporting point-in-time availability checks.

This evidence supports HAR-RV as a model-dependent prospective RV forecast layer. It does not support guaranteed forecasting superiority.

8. Regime Modelling

The regime modelling ladder is intentionally progressive. Each layer addresses a different modelling need.

Model layer	Purpose	Limitation
Threshold regimes	Simple interpretable baseline	Deterministic and threshold-sensitive
Gaussian HMM	Latent state classification	Does not directly model observed-series autocorrelation
Markov autoregression	AR-aware regime model	Reduced-form and numerically sensitive
MSVOL appendix	Volatility-regime robustness	Python-only MSVOL, not true R MSGARCH

8.1 Threshold regimes

Threshold regimes provide a deterministic benchmark. The inspected no-lookahead audit supports `uses_strict_prior_thresholds=True` across the audited rows: `True`.

Compact threshold-regime evidence:

market	state_name	n_days	fraction_days	avg_iv_ann	avg_vrp_har_gk
INDIA	calm	29.0000	0.0086	0.0227	0.0070
INDIA	transition	1735.0000	0.5124	0.0209	0.0024
INDIA	stress	1622.0000	0.4790	0.0440	0.0190
US	calm	55.0000	0.0066	0.0208	0.0134
US	transition	3828.0000	0.4591	0.0240	0.0129
US	stress	4455.0000	0.5343	0.0625	0.0365

8.2 Gaussian HMM

The Gaussian HMM estimates latent regimes from observed features. Filtered probabilities are used where time-safe probabilities are required. Full-sample smoothed probabilities are diagnostic only.

Compact HMM state evidence:

market	economic_state_name	n_observations	occupancy	mean_vrp_har_gk	mean_iv_ann	mean_rv_gk_22d_ann_lag1	hmm_model_valid
US	calm	3942.0000	0.4589	0.0093	0.0191	0.0064	True
US	transition	3391.0000	0.3948	0.0270	0.0444	0.0178	True
US	stress	1257.0000	0.1463	0.0700	0.1203	0.0648	True
INDIA	calm	2252.0000	0.6190	0.0024	0.0206	0.0093	True
INDIA	transition	1166.0000	0.3205	0.0210	0.0441	0.0207	True
INDIA	stress	220.0000	0.0605	0.0621	0.1164	0.0866	True

The HMM no-lookahead audits report `overall_passed=True` for both markets: US True, India True. The report therefore uses filtered-probability/no-smoothed-backtest wording.

8.3 Markov autoregression

Markov autoregression extends the regime ladder by allowing state-dependent autoregressive dynamics in the observed series. This addresses an important limitation of standard Gaussian HMM emissions.

Compact MAR state evidence:

market	economic_state_name	ar_lag1_phi	sigma2	persistence_prob	ar_stable	target_col
US	stress	0.9285	0.4259	0.9097	True	vrp_har_gk
US	calm	0.9587	0.0093	0.9641	True	vrp_har_gk
INDIA	stress	0.8660	0.3108	0.8103	True	vrp_har_gk
INDIA	calm	0.9970	0.0063	0.9593	True	vrp_har_gk

Both markets have two-state MAR summaries in the inspected files. The MAR no-lookahead audits report `passed=True`: US True, India True. MAR remains a reduced-form model and its regimes are economic interpretations, not observed ground truth.

8.4 MSVOL robustness appendix

The MSVOL layer is a Python-only Markov-switching volatility robustness appendix. It is not a true R MSGARCH implementation and is not used for strategy construction or backtesting.

Compact MSVOL evidence:

market	status	n_msvol_days	n_overlap_days	diagnostic_only	used_for_strategy	used_for_backtest
INDIA	ok	4204.0000	4204.0000	True	False	False
US	ok	9155.0000	9155.0000	True	False	False

The MSVOL no-lookahead audit reports 44 passed rows out of 44 audited rows.

9. Strategy Construction

The strategy layer converts regime and carry information into next-session exposure intentions.

The strategy outputs are not broker orders. They are research-layer exposure targets used for vectorised backtesting.

Compact strategy universe:

market	strategy_name	regime_model	available_fraction	mean_target_exposure	first_signal_observation_date	last_signal_observation_date
US	unconditional_full	unconditional	0.9999	-1.0000	1990-01-02	2026-05-14
US	threshold_hard_filter	threshold	0.9107	-0.4657	1990-01-02	2026-05-14
US	threshold_defensive	threshold	0.9107	-0.1214	1990-01-02	2026-05-14
US	hmm_prob_linear	gaussian_hmm	0.9999	-0.4586	1992-02-27	2026-04-14
US	hmm_prob_linear_carry	gaussian_hmm	0.9999	-0.4583	1992-02-27	2026-04-14
US	mar_prob_linear	markov_autoreg	0.9959	-0.6539	1992-02-27	2026-04-14
US	mar_prob_linear_carry	markov_autoreg	0.9959	-0.6513	1992-02-27	2026-04-14
INDIA	unconditional_full	unconditional	0.9998	-1.0000	2009-03-02	2026-05-15
INDIA	threshold_hard_filter	threshold	0.8054	-0.5210	2009-03-02	2026-05-15
INDIA	threshold_defensive	threshold	0.8054	-0.1367	2009-03-02	2026-05-15
INDIA	hmm_prob_linear	gaussian_hmm	0.9997	-0.6177	2011-05-17	2026-04-13
INDIA	hmm_prob_linear_carry	gaussian_hmm	0.9997	-0.3882	2011-05-17	2026-04-13

market	strategy_name	regime_model	available_fraction	mean_target_exposure	first_signal_observation_date	last_signal_observation_date
INDIA	mar_prob_linear	markov_autoreg	0.9959	-0.8014	2011-05-17	2026-04-13
INDIA	mar_prob_linear_carry	markov_autoreg	0.9959	-0.5601	2011-05-17	2026-04-13

The inspected table contains 7 US strategies and 7 India strategies. The signal no-lookahead audit records forward or diagnostic fields as present but excluded from strategy use where applicable; 22 audited rows have used_by_strategy=False.

10. Vectorised Research Backtest

The Phase 10 backtest is a vectorised research-proxy evaluation. It is not an executable option-chain simulation and does not report account returns.

The backtest evaluates proxy return units based on the project's VRP outcome construction and exposure intentions.

Important accounting caveats:

- Returns are research-layer proxy units.
- Cumulative curves are additive proxy curves, not account equity curves.
- No initial capital, margin, option contract sizing, or broker execution is modelled.
- Transaction costs are assumptions in proxy units.
- Overlapping 22-day labels make annualised metrics approximate.

Compact Phase 10 summary:

market	strategy_name	n_obs	total_return_proxy	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	hit_rate
INDIA	hmm_prob_linear	3638.0000	16.3937	1.1359	0.3207	3.5423	3.8483	-3.2562	0.6692
INDIA	hmm_prob_linear_carry	3638.0000	10.0665	0.6975	0.3193	2.1843	2.3633	-3.2562	0.4495
INDIA	mar_prob_linear	3638.0000	37.1641	2.5850	0.4502	5.7417	9.9048	-2.8145	0.8234
INDIA	mar_prob_linear_carry	3638.0000	30.2355	2.1030	0.4551	4.6210	8.0640	-2.7981	0.5722
INDIA	threshold_defensive	4204.0000	3.6845	0.2742	0.0671	4.0838	4.8435	-0.8432	0.4843
INDIA	threshold_hard_filter	4204.0000	14.0716	1.0473	0.2604	4.0224	4.6494	-3.3729	0.4843
INDIA	unconditional_full	4204.0000	83.3277	5.0212	0.6365	7.8888	13.3529	-6.1924	0.9469
US	hmm_prob_linear	8590.0000	41.9781	1.2316	0.1445	8.5253	13.1863	-1.5720	0.5124
US	hmm_prob_linear_carry	8590.0000	41.9610	1.2311	0.1445	8.5217	13.1809	-1.5720	0.5122

market	strategy_name	n_obs	total_return_proxy	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	hit_rate
US	mar_prob_linear	8590.0000	82.8569	2.4407	0.1985	12.2965	23.1696	-1.5019	0.6928
US	mar_prob_linear_carry	8590.0000	82.2909	2.4240	0.1987	12.2007	23.0167	-1.5019	0.6754
US	threshold_defensive	9156.0000	14.3671	0.4342	0.0450	9.6535	35.6028	-0.2496	0.4475
US	threshold_hard_filter	9156.0000	55.1127	1.6657	0.1662	10.0226	34.1459	-0.9984	0.4475
US	unconditional_full	9156.0000	226.4497	6.2476	0.5524	11.3094	27.8074	-5.3365	0.9508

All metrics above are research-proxy metrics. Annualized metrics are approximate because the observations are not independent daily returns; the metadata records annualization_periods=252 and research_proxy_not_trade_pnl=True.

The no-lookahead audit reports all rows passing: True. The Phase 10 final audit status is passed.

Metric-scoped comparison from backtest_summary.csv: by sharpe, the highest US row is mar_prob_linear at 12.2965 and the highest India row is unconditional_full at 7.8888. This is a research-proxy table comparison only, not a live-trading or account-return claim.

Selected final-report figures:

 US research-proxy cumulative curve on the common-start sample; not an account equity curve.

US research-proxy cumulative curve on the common-start sample; not an account equity curve.

 India research-proxy cumulative curve on the common-start sample; not an account equity curve.

India research-proxy cumulative curve on the common-start sample; not an account equity curve.

 US research-proxy drawdown diagnostic; not account drawdown.

US research-proxy drawdown diagnostic; not account drawdown.

 India research-proxy drawdown diagnostic; not account drawdown.

India research-proxy drawdown diagnostic; not account drawdown.

11. Robustness Checks

The robustness layer tests whether Phase 10 findings are sensitive to assumptions and sample choices.

Robustness evidence availability:

Robustness area	Evidence	Inspected status
Cost sensitivity	reports/tables/phase_10/robustness_cost_sensitivity.csv	70 rows across cost-bps scenarios 0.0, 10.0, 2.5, 20.0, 5.0
Subperiod behaviour	reports/tables/phase_10/robustness_subperiods.csv	70 rows across subperiods COVID, China_Devaluation_Vol_Shock, EuroDebt_US_Downgrade, GFC, RateShock, TaperTantrum, Volmageddon
Crisis windows	reports/tables/phase_10/crisis_window_performance.csv	70 rows across windows COVID, China_Devaluation_Vol_Shock, EuroDebt_US_Downgrade, GFC, RateShock, TaperTantrum, Volmageddon
Tradable proxy detection	reports/tables/phase_10/tradable_proxy_detection.json	status skipped

The tradable proxy detection artifact reports status skipped with reason: required tradable proxy data not found; Phase 10 does not download new tradable proxy data.

These are sensitivity diagnostics over research-proxy assumptions. They do not prove robustness to all future execution costs, crises, or market conditions.

12. Cross-Market US-India Analysis

Phase 13 adds a cross-market analysis layer. It is analysis-only and does not alter the locked Phase 9 strategy universe.

The cross-market layer includes same-date descriptive diagnostics, lagged-US predictive/statistical diagnostics, Granger-style lead-lag diagnostics, logistic incremental-signal tests, and an analysis-only India overlay.

Alignment and no-lookahead summary:

model	n_india_dates	n_rows	n_same_date_violations	n_same_date_or_future_us_violations	passes_no_lookahead
gaussian_hmm	3638.0000	3638.0000	0.0000	0.0000	True
markov_autoreg	3638.0000	3638.0000	0.0000	0.0000	True

Logistic model comparison:

model	local_n_obs	plus_us_n_obs	local_auc	plus_us_auc	delta_auc	local_pseudo_r2	plus_us_pseudo_r2	delta_pseudo_r2	likelihood_ratio_p_value
gaussian_hmm	3637.0000	3637.0000	0.9959	0.9968	0.0009	0.8975	0.9005	0.0031	0.2783
markov_autoreg	3637.0000	3637.0000	0.9050	0.9002	-0.0048	0.3517	0.3689	0.0172	0.0000

India overlay summary:

model	strategy	cutoff	n_obs	base_sharpe	overlay_sharpe	base_max_drawdown	overlay_max_drawdown	analysis_only
gaussian_hmm	hmm_prob_linear_carry	0.5000	514.0000	0.8958	0.7773	-0.7231	-0.7233	True
gaussian_hmm	hmm_prob_linear_carry	0.6000	514.0000	0.8958	0.7773	-0.7231	-0.7233	True
gaussian_hmm	hmm_prob_linear_carry	0.7000	514.0000	0.8958	0.7773	-0.7231	-0.7233	True

model	strategy	cutoff	n_obs	base_sharpe	overlay_sharpe	base_max_drawdown	overlay_max_drawdown	analysis_only
markov_autoreg	mar_prob_linear_carry	0.5000	562.0000	4.8201	3.7103	-0.5531	-0.5538	True
markov_autoreg	mar_prob_linear_carry	0.6000	562.0000	4.8201	3.9909	-0.5531	-0.5538	True
markov_autoreg	mar_prob_linear_carry	0.7000	562.0000	4.8201	4.0979	-0.5531	-0.5538	True

The Phase 13 run-status artifact reports status ok. Same-date diagnostics are descriptive only. Lagged-US and logistic diagnostics are predictive/statistical diagnostics only. The India overlay is analysis-only and outside the locked Phase 9 strategy universe.

Selected final-report figures:

US-India VRP descriptive cross-market diagnostic; not causal evidence.

US-India VRP descriptive cross-market diagnostic; not causal evidence.

US-India stress probability descriptive cross-market diagnostic; not causal evidence.

US-India stress probability descriptive cross-market diagnostic; not causal evidence.

Lagged-US versus India stress predictive diagnostic; not causal proof.

Lagged-US versus India stress predictive diagnostic; not causal proof.

13. IBKR Paper-Signal Readiness Appendix

Phase 11 is an operational readiness appendix. It converts research signals into guarded paper-signal artifacts and validates that live-order pathways remain blocked.

Allowed wording:

paper-signal readiness layer
order-guard validation
configuration and risk-check demonstration
no broker orders placed

Forbidden wording:

paper trading results
execution layer

live strategy
broker backtest

Compact guard evidence:

artifact	field	value
live_order_guard_report.json	passed	True
live_order_guard_report.json	violations	[]
phase11_integration_report.json	passed	True
phase11_integration_report.json	violations	[]
run_metadata.json	live_order_sent	False
run_metadata.json	live_orders_enabled	False
run_metadata.json	allow_order_placement	False
run_metadata.json	paper_only	True
run_metadata.json	kill_switch	True
broker_metadata.json	live_order_sent	False
broker_metadata.json	live_orders_enabled	False
broker_metadata.json	allow_order_placement	False
broker_metadata.json	paper_only	True
broker_metadata.json	kill_switch	True
risk_check_report.csv	data_rows	0
risk_check_report.csv	columns	valid risk-check schema

The risk-check CSV has zero data rows but a valid risk-check schema. Broker-sensitive details are excluded from this report. The inspected guard and metadata fields support the statement that no broker orders were placed: True.

14. Main Findings

This section is based only on inspected local evidence listed in the Phase 14 evidence review and claim audit.

Finding 1 - VRP construction validity

The project constructs aligned US and India VRP panels using VIX/India VIX implied-variance proxies and daily OHLC realised-variance proxies. The VRP metadata identifies `rv_gk_22d_ann` as the primary realised-variance column, `vrp_backward_gk` as the primary backward VRP column, and `vrp_forward_expost_gk_label` as an evaluation outcome label.

Finding 2 - HAR forecast usefulness

The HAR-RV layer provides a model-dependent, point-in-time realised-variance forecast for prospective VRP construction. The forecast accuracy table contains HAR and baseline forecast rows for both markets, and the audit table records forecast availability and blocked rows by reason.

Finding 3 - Regime interpretability

The regime ladder provides interpretable threshold, HMM, and MAR state summaries for both markets. The HMM no-lookahead audits report `overall_passed=True` for both markets, and the MAR no-lookahead audits report `passed=True` for both markets. MAR adds an AR-aware reduced-form layer beyond the Gaussian HMM.

Finding 4 - Research-proxy strategy performance

In `backtest_summary.csv`, Phase 10 contains seven research-proxy strategy rows per market. By the metric `sharpe`, the highest US row is `mar_prob_linear` at 12.2965 and the highest India row is `unconditional_full` at 7.8888. This comparison is metric-scoped and applies only to the vectorised research-proxy table.

Finding 5 - Robustness and cost sensitivity

The robustness artifacts provide sensitivity diagnostics across cost-bps assumptions, named subperiods, and crisis windows. The tradable proxy detection artifact reports status `skipped`, so the report should not present Phase 10 as true instrument-level tradable proxy evidence.

Finding 6 - Cross-market evidence

Phase 13 alignment and no-lookahead tables report zero same-date or future-US violations for the inspected models. Logistic comparison tables and overlay diagnostics are available, but they support only predictive/statistical wording and analysis-only overlay wording, not causal conclusions.

Finding 7 - Paper-signal readiness

The Phase 11 guard artifacts report passing live-order guard and integration checks, with `live_order_sent=False` in both run and broker metadata. This supports paper-signal readiness and live-order guard wording; no broker orders were placed.

15. Terminology Lock

The report uses the following terms in a restricted way.

Term used in report	Meaning	Not meant as
Research-proxy return	Additive VRP proxy backtest unit	Account return
Exposure intention	Signal target from the strategy layer	Broker order
Filtered probability	Time-t available regime probability	Full-sample smoothed probability
Cross-market predictive diagnostic	Statistical lead-lag or predictive association test	Causal transmission proof
MSVOL robustness	Python Markov-switching volatility check	True R MSGARCH
Paper-signal readiness	Guarded signal-format and risk-check demonstration	Broker execution or paper trading result

16. Limitations

The full final limitations document is:

```
reports/final/limitations.md
```

Core limitations:

- VIX and India VIX are implied-volatility proxies, not variance swap quotes.
- Daily OHLC realised variance estimators are proxies for true realised variance.
- A 22-trading-day horizon approximates a one-month horizon but is not identical to 30 calendar days.
- Forward ex-post VRP labels are outcomes only, not tradable features.
- HAR forecasts are model-dependent and constrained by training-window choices.
- Gaussian HMM does not directly model observed-series autocorrelation.
- Markov autoregression addresses observed-series autocorrelation but remains reduced-form.
- MSVOL is Python-only volatility-regime robustness, not true R MSGARCH.

- Strategy outputs are exposure intentions, not broker orders.
- Phase 10 returns are research-layer proxy units, not account returns.
- Overlapping 22-day outcome labels make annualised metrics approximate.
- Phase 11 did not place broker orders.
- Phase 12 paper execution was skipped and remains future optional work.
- Phase 13 diagnostics are statistical/predictive diagnostics, not causal proof.
- Generated panels remain local and are not committed to GitHub.

17. Future Work

The full future work document is:

```
reports/final/future_work.md
```

Future extensions:

- True option-chain PnL using historical option chains.
- Instrument-level execution modelling with bid/ask spreads, liquidity, margin, and contract selection.
- Optional IBKR paper execution adapter.
- True R MSGARCH implementation.
- Intraday realised variance using high-frequency data.
- Broader international volatility-index comparison.
- Production-grade monitoring and deployment safeguards.

These are explicitly outside the current EPAT submission scope.

18. Reproducibility

The full reproducibility documentation is:

```
docs/reproducibility.md
reports/final/reproducibility_note.md
docs/commands.md
```

A reviewer can inspect the tracked repository without receiving large local data files. The reproducible tracked layer includes:

```
source code
configs
scripts
tests
docs
README files
environment template
directory placeholders
final report package
```

Local-only generated artifacts include:

```
raw market data
processed feature panels
trained model outputs
regime panels
strategy signal panels
backtest panels
broker/paper-signal cache
cross-market panels
```

Minimal validation commands:

```
pip install -e .
pytest
python scripts/download_data.py --dry-run
```

Final release validation commands are documented in:

```
docs/release_checklist.md
docs/commands.md
```

19. Appendix A — Report Artifact Map

Report area	Evidence inventory
Claims audit	reports/final/result_claims_audit.md
Table inventory	reports/final/table_inventory.md
Figure inventory	reports/final/figure_inventory.md
Selected artifacts	reports/final/selected_artifacts.md
Release checklist	docs/release_checklist.md
Submission map	docs/submission_package.md

20. Appendix B — Required Final Validation

Before freezing Phase 14, run:

```
git diff --check
git status --short
git ls-files | findstr /i "\.parquet \.pkl \.pickle \.joblib \.pt \.pth \.log \.env"
git ls-files | findstr /i "broker_cache data/raw data/interim data/processed"
pytest
python scripts/validate_phase11.py --print-json
python scripts/run_cross_market_analysis.py --validate-inputs-only
pytest tests/test_cross_market_alignment.py tests/test_cross_market_no_lookahead.py tests/test_cross_market_stats.py tests/test_cross_market_overlay.py
tests/test_phase13_artifact_mutation.py tests/test_phase13_datetime_dtype.py
```

Expected policy result:

No heavy generated artifacts tracked.
No credentials tracked.
No broker-sensitive artifacts tracked.
No live-order evidence.
No unaudited numeric report claims.

21. Appendix C — Final Report Completion Checklist

Before PDF export:

- ☐ All numeric placeholders resolved or explicitly left as placeholders.
- ☐ Every major claim appears in `result_claims_audit.md`.
- ☐ Every table appears in `table_inventory.md`.
- ☐ Every figure appears in `figure_inventory.md`.
- ☐ All figures have research-proxy or diagnostic captions.
- ☐ No account-return wording.
- ☐ No live-trading profitability wording.
- ☐ No true option-chain PnL wording.
- ☐ No causal cross-market wording.
- ☐ No true MSGARCH wording.
- ☐ Phase 11 is appendix-only.
- ☐ Phase 12 is skipped / future optional.
- ☐ Limitations are complete.
- ☐ Future work is complete.
- ☐ Reproducibility note is complete.
- ☐ PDF visually inspected after export.